PORTALDOT HACKER HOUSE 2026 · TRACK 4, ECOSYSTEM DAPPS

Allowance

Give your AI agent an allowance, not a wallet.

Spending rules for AI agents, enforced by a contract on Portaldot V3.

Ozan Turan · github.com/Oziman13/allowance · oziman13.github.io/allowance

THE PROBLEM

Today, giving an AI agent money means trusting it with all of it.

A wallet key

Can spend everything in the wallet.

A proxy

Limits which calls it makes, not how much it spends.

A confirm button

Lives in the app, so a compromised agent can skip it.

One bad prompt, a hallucinated invoice or a prompt-injected tool, and the funds are gone. Signing every payment by hand removes the point of having an agent.

HOW IT WORKS

Rules in, payments out, the rest waits.

1

Set the rules

The owner opens an allowance for an agent key: a budget in POT or any ERC-20, a cap per payment, a cap per day, week or month, trusted recipients, an expiry.

2

The agent pays

Inside the rules, payments settle at once. The agent’s gas comes from the same budget, under the same caps.

3

The owner decides the rest

A payment outside the rules is held with the agent’s reason. Approve, approve and trust, or reject. One transaction freezes the agent.

THE GUARANTEE

If the agent’s key leaks, the thief is still bound by the daily cap.

The rules live in the contract, not in a server or an app. There is no off-chain signer to compromise. A 1,000-run fuzz test checks it: money that leaves without the owner’s approval never exceeds the period limit.

PROOF · PORTALDOT V3 TESTNET · OCT 23, 2026

A real AI assistant, Claude, paying through Allowance

Block	What happened	Cost
159822	Claude pays 0.3 POT for a market-data API: paid at once	0.032 POT
160197	Claude asks for 2 POT, over the 1 POT cap: held, not paid	19.24 POT
160213	Owner approves Claude’s request: 2 POT paid	0.019 POT
160231	Owner freezes the agent; Claude then refuses to pay	3.86 POT
160258	Owner unfreezes the agent; the slot’s deposit returns	0 POT

Asked to pay again without an address, Claude asked for the recipient instead of guessing. The MCP server holds only the agent key.

Contract 0xdF8567f1fA9F6aFBe1c8f99453544C35009ac07f · every hash in docs/PROOF.md

MEASURED ON CHAIN

Cheap to use, tested where it counts

0.032

POT for an instant payment: it only updates existing storage

0.019

POT for the owner to approve a held payment

779

POT to deploy, down from about 1,047 with a size-tuned build

33

tests on every push: 27 contract, incl. a 1,000-run fuzz, and 6 MCP end-to-end

WHY PORTALDOT

Agent payments are on-chain activity Portaldot wants

Plain Solidity on V3

The Foundry build runs unchanged on pallet-revive. The code on chain is byte for byte the repository's.

POT is the budget and the gas

Agents spend and refuel in POT, under the same caps, so every payment is native activity.

A guide for the next builders

"Getting started with Solidity on Portaldot V3": funding, storage deposits, errors, pruned history. All measured.

Usable by any AI assistant

The MCP server brings Claude, Cursor and other MCP clients to Portaldot with five tools.

WHAT'S BUILT

A working product, not a mock-up

Part	What it does
Contract	Caps, periods, trusted recipients, expiry, freeze, key rotation, held requests
Owner dashboard	Spending meter, inbox for held payments, controls, activity with tx hashes
Simulation mode	The same bytecode in an in-browser EVM: try every rule without a wallet
MCP server	Five tools and an agent skill: preview first, never split a payment
Builder guide	Getting started with Solidity on Portaldot V3, measured on chain
Check script	Re-checks the deployed code and every proof transaction from the RPC

BUSINESS MODEL

Open protocol, paid team layer, no token

FREE

Open and MIT

The contract, dashboard and MCP server, for anyone to use and build on.

SUBSCRIPTION

Teams

Alerts when a payment waits, approvals by more than one member, spending reports, hosted agent runners.

ON MAINNET

Protocol fee

A v2 contract adds a small, capped fee on instant payments. The testnet contract has none.

Revenue grows with agent payments on Portaldot, the on-chain activity the ecosystem rewards.

ROADMAP

From testnet to Portaldot-native

Done

- Deployed and exercised on V3 testnet
- Real AI assistant paying via MCP
- Dashboard, simulation, builder guide

By the final, Nov 6

- Owners manage from a Portaldot (Substrate) wallet
- MCP server published on npm
- Demo video

After

- Allowances in Portaldot Assets tokens
- v2: resolved requests return their deposit
- Alerts for held payments; the teams layer

WHY KEEP SUPPORTING IT

AI agents will pay for things. On Portaldot, they can do it within rules.

Every allowance is a stream of small POT payments. Support takes it to mainnet: a security review of the contract, Portaldot-native owners and Assets-token budgets, and an open SDK so other Portaldot apps can give their agents allowances.

Try it: oziman13.github.io/allowance Code: github.com/Oziman13/allowance

Ozan Turan · solo builder · Telegram @oziman1 · Testnet only; not audited